

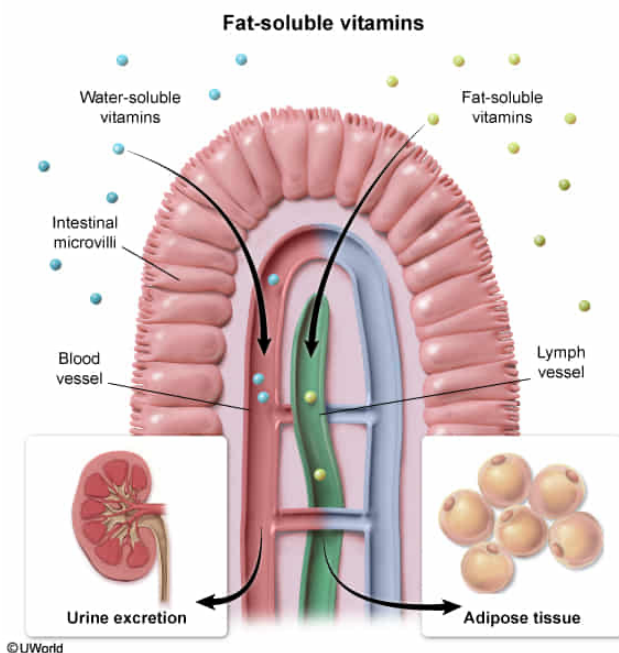
Which type of molecule, primarily obtained from the diet, would most likely accumulate in adipose tissue?

- ☐ A. Steroid hormones [6%]
- ☐ B. Water-soluble vitamins [3%]
- ☐ C. Peptide hormones [1%]
- ✓ ☒ D. Fat-soluble vitamins [89%]

Correct

88% answered correctly

### Explanation:



The human body is capable of using precursor molecules to synthesize many of the biomolecules needed for proper function. However, humans lack the enzymes necessary to synthesize certain molecules such as the **essential amino and fatty acids**, and the class of molecules known as **vitamins**. These molecules **must be obtained from the diet**.

Vitamins are classified as either **water-soluble** (B series, C) or **fat-soluble** (A, D, E, K). **Water-soluble vitamins** contain several hydrophilic groups and are easily dissolved in aqueous environments. They are not stored in the body, and the excess is **excreted in the urine (Choice B)**. For this reason, water-soluble vitamins must be frequently replenished.

In contrast, **fat-soluble vitamins** are highly hydrophobic. They dissolve in lipid rather than aqueous environments and cannot be easily excreted in urine.

Instead, the excess is stored and **accumulates in adipose tissue**, which consists mainly of fat cells.

**(Choice A)** **Steroid hormones** are produced primarily in the adrenal cortex and the gonads from cholesterol and other precursors. They are not required in the diet and do not generally accumulate in adipose tissue.

**(Choice C)** **Peptide hormones** are readily synthesized by the body, and are not required in the diet. They are generally soluble in water, and are not stored in adipose tissue.

**Educational objective:**

Certain biomolecules such as vitamins and the essential amino and fatty acids cannot be synthesized in the human body and must be obtained through the diet. Water-soluble vitamins (B series, C) are excreted in the urine whereas fat-soluble vitamins (A, D, E, and K) are stored in adipose and other fatty tissues.

**Time spent:**0 sec

**Subject:**

Biochemistry

**QID:**402074

**System:**

Carbohydrates, Nucleotides, and Lipids